Tuesday 30th November 2021

Complete the worksheet

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DO NOW

Radioactive decay

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Learning Objective

- Describe radioactive decay
- Describe the types of nuclear radiation
- Understand the processes of alpha decay and beta decay

Radioactive decay:

- Most atoms are stable but some are not
- If the atom has an unstable nucleus it will undergo radioactive decay to become more stable
- An atom with an unstable nucleus is called radioactive

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Radioactive decay:

- **The activity of a radioactive atom is measured by how many nuclear decays will happen every second**
- This is measured in becquerels (Bq)
- 1 Bq = 1 decay per second

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Radioactive decay:

- If the activity of a radioactive source is 10 Bq, how many counts would be recorded in 30 seconds?

Radioactive decay:

- Worksheet time! (activity)

Radioactive decay:

- Radioactivity is a **random** process and so you cannot predict when the nucleus will decay
- Can you predict when corn kernels will start to pop?

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Radioactive decay:

- There are 3 main types of nuclear radiation
  - Alpha
  - Beta
  - Gamma
- Neutron decay is where high-speed neutrons are released from the nucleus

Name the 3 types of nuclear radiation:

The 3 main types of nuclear radiation are:

- Alpha
- Beta
- Gamma

Alpha decay:

- In alpha decay an **alpha particle** is emitted from the nucleus
- An alpha particle is the same a **helium nucleus**
- What is the number of protons and neutrons in a helium nucleus?
- 2 protons and 2 neutrons

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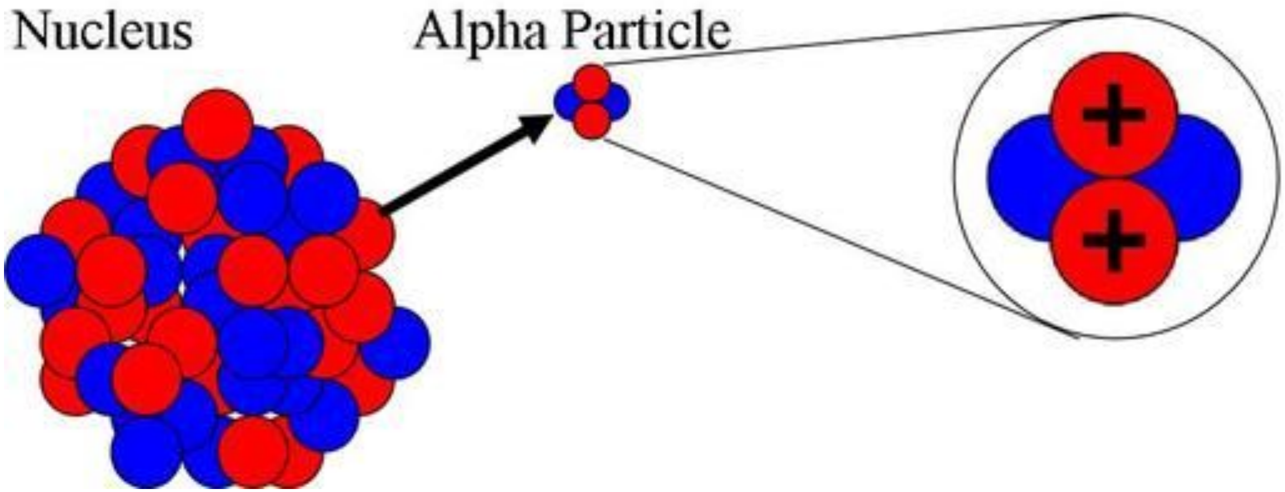
- During alpha decay an atom’s nucleus decays, it loses two protons and two neutrons

The number of protons has changed, so the decayed atom has changed into a **new element**.

Alpha decay:



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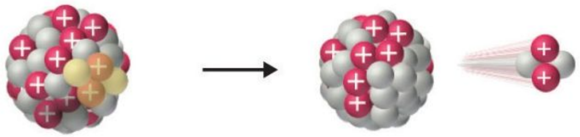
Alpha decay:

$$^{239}_{94}\mathrm{Pu} \rightarrow ^{235}_{92}\mathrm{U} + ^{4}_{2}\alpha$$

Alpha decay:

Decay Type	Radiation Emitted	Generic Equation	Model
Alpha decay	${}^{4}_{2}\alpha$	${}^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}X' + {}^{4}_{2}\alpha$	 Parent → Daughter Alpha Particle

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Alpha decay:

- Write down what alpha decay is:
- Write down what an alpha particle is:
    - Alpha decay is when an ………………….. is emitted from the …………………..
    - An alpha particle consists of ….. Protons and …….. neutrons

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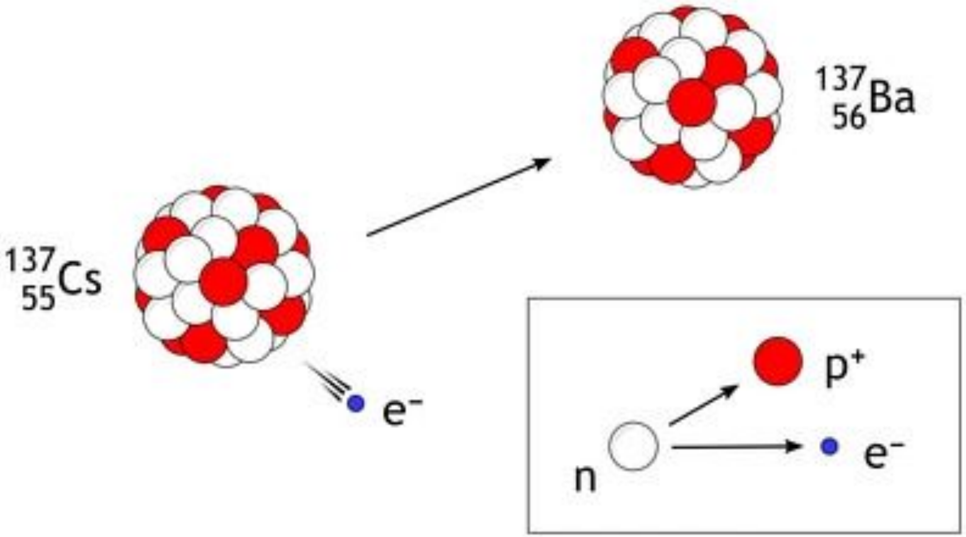
Beta decay:

- In beta decay **one of the neutrons** in the nucleus decays into a **proton and an electron**
- The **electron** is emitted from the nucleus and is called a **beta particle**

Beta decay:



What changes have happened to the nucleus of **Cs**?

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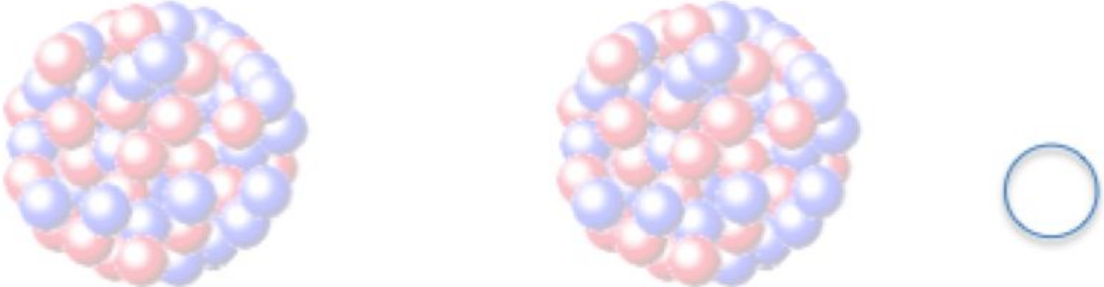
Beta decay:

- When **beta decay** happens:
- The nucleus has 1 more proton, so the atomic number increases by 1
- The nucleus has 1 less neutron, the mass number is unchanged
- A new element is formed

Beta decay:



$$^{14}_{6}\mathrm{C} \longrightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\beta$$

What is the difference between the Carbon and Nitrogen nucleus?

Gamma decay:

- In gamma decay **gamma rays** are emitted from the nucleus
- They are very **high-energy electromagnetic waves**
- They have no charge and no mass
- Gamma decay **does not** cause a change in the nucleus of the atom

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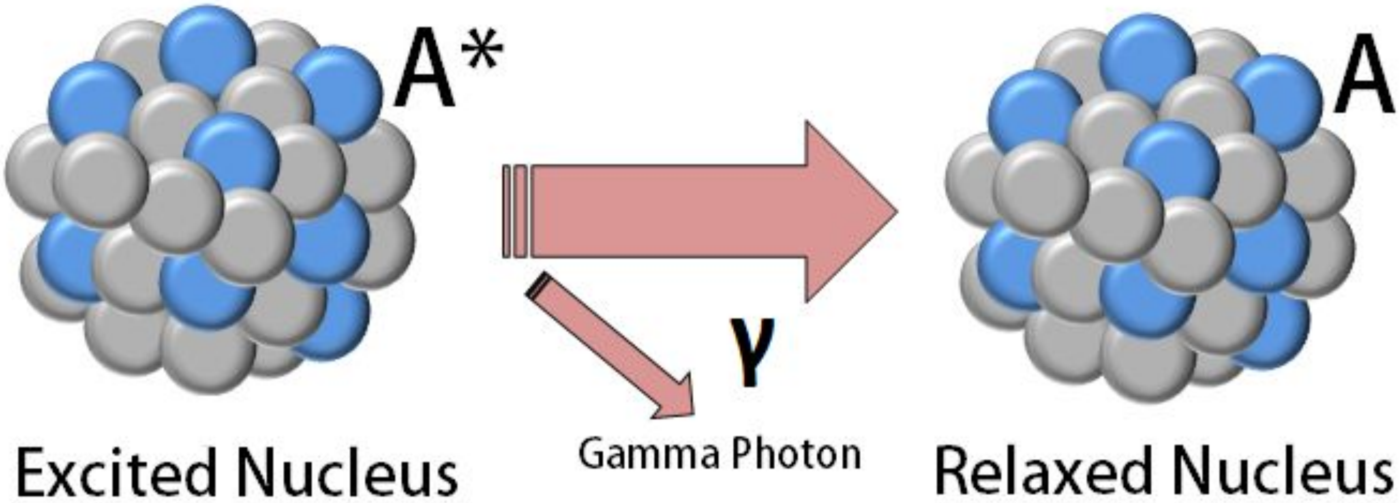
Gamma decay:



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Gamma decay:

- In gamma decay ……………………. are emitted. These are very high …………. electromagnetic waves. This decay does not cause a change in the ………………